Lab 8

AP25110010807

S.Jignesh

Que1 code:

main.cpp

```cpp

#include <iostream>
#include <queue>
#include <vector>
#include <string>
using namespace std;

struct Node {
    char letter;
    int frequency;
    Node *left, *right;

    Node(char ch, int f) {
        letter = ch;
        frequency = f;
        left = right = nullptr;
    }
};

struct Compare {
    bool operator()(Node* a, Node* b) {
        return a->frequency > b->frequency;
    }
};

void printCodes(Node* root, string code, int &totalBits) {
    if (root == nullptr)
        return;

    if (root->left == nullptr && root->right == nullptr) {
        cout << root->letter << "\t"
             << root->frequency << "\t"
             << code << endl;

        totalBits += root->frequency * code.length();
        return;
    }

    printCodes(root->left, code + "0", totalBits);
    printCodes(root->right, code + "1", totalBits);
}

```

```cpp

int main() {
    char letters[] = {'a', 'c', 'd', 'e', 'o', 'm', 's', 't', 'u'};
    int freq[] = {20, 11, 2, 10, 15, 8, 10, 22, 2};

    priority_queue<Node*, vector<Node*>, Compare> pq;

    for (int i = 0; i < 9; i++) {
        pq.push(new Node(letters[i], freq[i]));
    }

    // Build the Huffman tree
    while (pq.size() > 1) {
        Node* left = pq.top();
        pq.pop();

        Node* right = pq.top();
        pq.pop();

        Node* parent = new Node('-', left->frequency + right->frequency);
        parent->left = left;
        parent->right = right;

        pq.push(parent);
    }

    Node* root = pq.top();

    int totalBits = 0;

    cout << "Letter\tFrequency\tHuffman Code\n";
    cout << "---------------------------------\n";

    printCodes(root, "", totalBits);

    cout << "\nTotal encoded bits = " << totalBits * 100;
    cout << "\nMemory required = " << (totalBits * 100 + 7) / 8
         << " bytes\n";

    return 0;
}

```

Output:

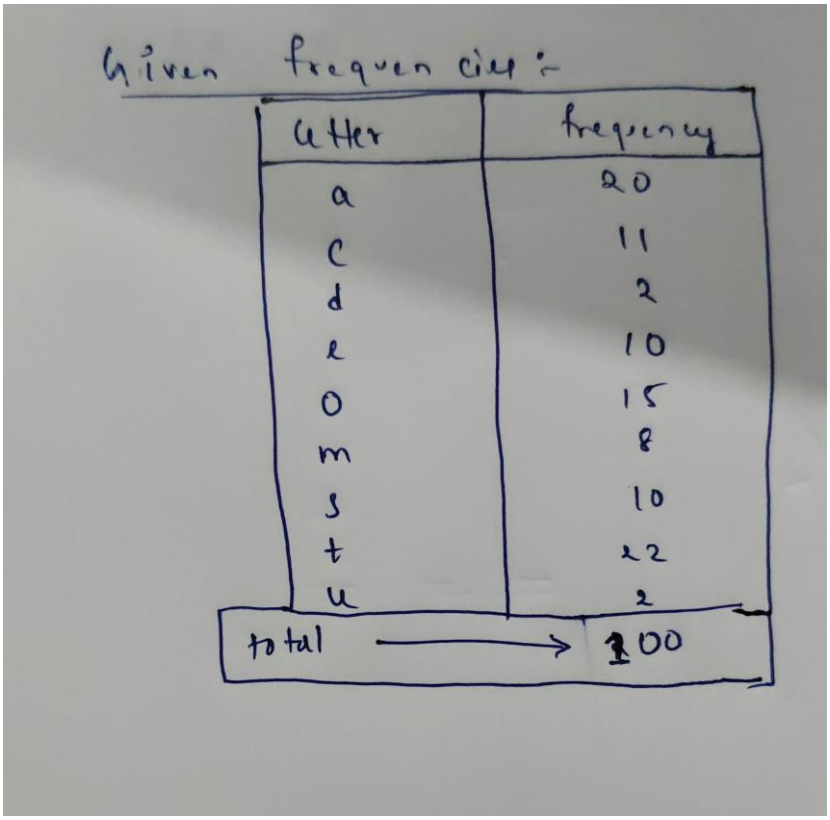
```
Letter  Frequency       Huffman Code
---------------------------------
a       20      00
t       22      01
c       11      100
d       2       10100
u       2       10101
m       8       1011
o       15      110
s       10      1110
e       10      1111

Total encoded bits = 29400
Memory required = 3675 bytes


...Program finished with exit code 0
Press ENTER to exit console.
```

Que2:

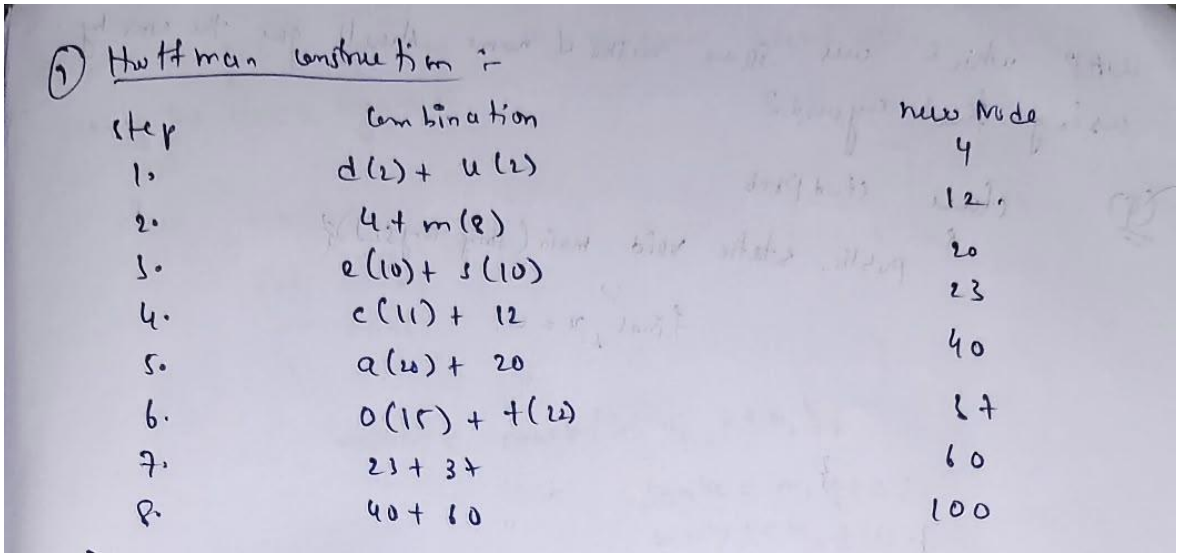
a.



Given frequencies :-

Letter	frequency
a	20
c	11
d	2
e	10
o	15
m	8
s	10
t	22
u	2
total →	100

1.

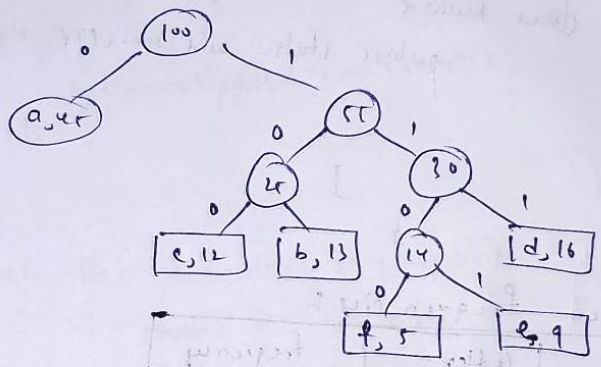


(i) Huffman construction :-

step	combination	new Node
1.	d(2) + u(2)	4
2.	4 + m(8)	12
3.	e(10) + s(10)	20
4.	c(11) + 12	23
5.	a(20) + 20	40
6.	o(15) + t(22)	37
7.	23 + 37	60
8.	40 + 60	100

2.

Diagram:-



(a) Huffman encoded table:-

Letter	freq	Huffman code
a	20	00
b	11	100
d	2	10100
e	10	010
o	15	110
m	8	1011
s	10	011
t	22	111
u	2	10101

3.

(a) memory required:-

Letter	freq	Code length	total bits
a	20	2	40
e	11	3	33
d	2	5	10
e	10	3	30
o	15	3	45
m	8	4	32
s	10	3	30
t	22	3	66
u	2	5	10
total	100		296

total bits = $296 \times 100 = 29600$

memory = $\frac{29600}{8} = 3700$ bytes